

Challenge Write-Up Template

Instructions: Copy this template for every challenge you submit. Fill in every section — a write-up that only shows the final flag without explanation will not receive full credit. Delete these italicized instruction lines before submitting.

Challenge Information

Student Name	Daniella Marie A. Leonud
Section / Class	BSIT-3G
Challenge Name	Piece by Piece
Category	General Skills
Points / Difficulty	Easy
Date Solved	August 9, 2026
Flag Format Given	picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_da494d2e}

1. Challenge Description

Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copy-paste the original prompt.

In this Piece by Piece challenge, I was given a situation where a ZIP file had been divided into several smaller pieces. My task was to combine all of the pieces back into one complete ZIP file and then extract the file inside it to find the hidden flag. The archive was also password-protected, so after combining the pieces, I needed to use the password provided in the instructions to extract its contents.

2. Initial Analysis / Reconnaissance

What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?

First, I opened the challenge page and launched the instance in the workspace. After starting the instance, I needed to connect to the remote Linux environment using SSH. Once connected, I checked the contents of the current directory using **ls -la**. The command showed several files, including **instructions.txt** part_aa, part_ab, part_ac, part_ad, part_ae. I opened instructions.txt using cat **instructions.txt** to understand what I needed to do next.

3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

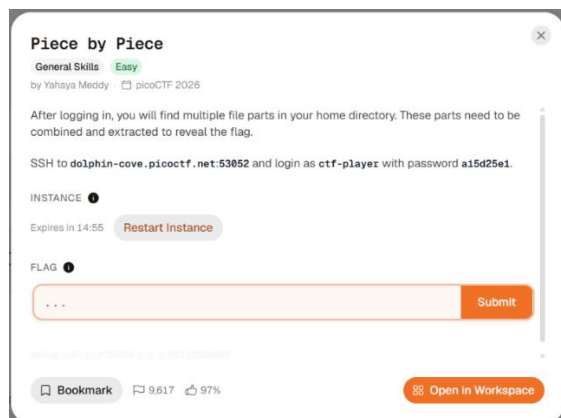
Tool / Command	What it was used for	Result / Output
ssh ctf-player@dolphin-cove.picocf.net -p 53052	Connected securely to the remote challenge server environment via SSH	Successfully logged into the Ubuntu Linux terminal session.
ls -la	Listed all files and directories, including hidden ones, with detailed permissions and file sizes.	Revealed instructions.txt and five split parts (part_aa to part_ae).
cat instructions.txt	Displayed the text contents of the instructions file to read the challenge hints.	Showed hints mentioning the split zip file, the password "supersecret", and instructions to recombine them.
cat part_* > flag.zip	Combined all files starting with the name part_ into a single unified zip file named flag.zip.	Successfully generated the complete compressed archive file named flag.zip.
unzip -P supersecret flag.zip	Extracted the contents of the password-protected flag.zip file using the password supersecret.	Extracted flag.txt into the directory.
cat flag.txt	Displayed the contents of the extracted flag.txt file	Printed the full challenge flag to the screen.
Gemini AI	Gave step-by-step guidance and helped solve terminal errors.	Guided me through the entire challenge.

4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

Step 1

First, I opened and read the challenge description. Then I launch instance and open it in the workspace.



Step 2

I connected to the provided remote instance via SSH using the provided command (**ssh ctf-player@dolphin-cove.picocftf.net -p 53052**) in the workspace terminal. And also I was asked before continuing to connect. And lastly, I need to type the password before confirming the connection.

```
leonuddaniella-academy@webshell:~$ ssh ctf-player@dolphin-cove.picocftf.net -p 53052
The authenticity of host '[dolphin-cove.picocftf.net]:53052 ([3.13.34.175]:53052)' can't be established.
ED25519 key fingerprint is SHA256:rdvWhF7Klwlu1EivysxTh/FjeqFI0dSEK5gaelaw9t8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '[dolphin-cove.picocftf.net]:53052' (ED25519) to the list of known hosts.
ctf-player@dolphin-cove.picocftf.net's password:
```

Step 3

After confirming the connection host key, I successfully entered the Linux environment.

```
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 7.0.0-1009-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Step 4

Once inside the terminal, I executed **ls -la** to inspect the directory contents. I noticed five split file chunks (part_aa, part_ab, part_ac, part_ad, part_ae) alongside an instructions.txt file. I ran **cat instructions.txt**. This displayed the information provided by the challenge. The instructions explained that the separate pieces needed to be combined back into a single ZIP archive. It also provided the password **supersecret**. In this part, I understood that I first needed to reconstruct the ZIP file before I could extract anything from it.

```
ctf-player@pico-chall$ ls -la
total 28
drwxr-xr-x 1 ctf-player ctf-player 20 Aug  9 06:36 .
drwxr-xr-x 1 root      root        24 Feb  4 2026 ..
drwx----- 2 ctf-player ctf-player 34 Aug  9 06:36 .cache
-rw-r--r-- 1 root      root        67 Feb  4 2026 .profile
-rw-r--r-- 1 ctf-player ctf-player 282 Feb  4 2026 instructions.txt
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4 2026 part_aa
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4 2026 part_ab
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4 2026 part_ac
-rw-r--r-- 1 ctf-player ctf-player 51 Feb  4 2026 part_ad
-rw-r--r-- 1 ctf-player ctf-player 35 Feb  4 2026 part_ae
ctf-player@pico-chall$ cat instructions.txt
Hint:
- The flag is split into multiple parts as a zipped file.
- Use Linux commands to combine the parts into one file.
- The zip file is password protected. Use this "supersecret" password to extract the zip file.
- After unzipping, check the extracted text file for the flag.
```

Step 5

To join all the pieces back, I used the cat command (**cat part_* > flag.zip**). This took all the files starting with part_ in alphabetical order and concatenated their raw data into a new file named **flag.zip**. With the full flag.zip file created, I needed to unpack it. Since the instructions mentioned it was password-protected, I **ran unzip -P supersecret flag.zip**. The -P flag allowed me to pass the password directly in the command line. Finally, I displayed the flag by typing **cat flag.txt**.

```
ctf-player@pico-chall$ cat part_* > flag.zip
ctf-player@pico-chall$ unzip -P supersecret flag.zip
Archive:  flag.zip
  extracting: flag.txt
ctf-player@pico-chall$ cat flag.txt
ctf-player@pico-chall$ Connection to dolphin-cove.picoctf.net closed by remote host.
Connection to dolphin-cove.picoctf.net closed.
```

Step 6

Finally, I copied the flag from the terminal and pasted it into the submission box .The flag was accepted as correct, which confirmed that I had completed the Piece by Piece challenge.

Piece by Piece
General Skills Easy
by Yahaya Meddy · 📅 picoCTF 2026

[Launch Instance](#) Additional details appear once you launch your instance.

FLAG ⓘ

Correct flag!

5. Flag

```
picoCTF{z1p_and_spl1t_f1l3s_4r3_fun_da494d2e}
```

6. Key Concept / What I Learned

In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.

This challenge demonstrated how I can split large files into smaller pieces for easier transfer or storage, and how I can seamlessly put them back together in order using Linux stream redirection tools like cat. Also, it helps me understand the basic archive handling in Linux, specifically how to extract password-protected zip archives via the command line using the unzip utility.

7. References

List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.

Linux Command Line Documentation (https://www.gnu.org/software/coreutils/manual/html_node/cat-invocation.html)

Zip / Unzip Documentation (<https://linux.die.net/man/1/unzip>)

picoCTF (<https://picoctf.org/>)

Gemini AI (<https://gemini.google.com/>)
